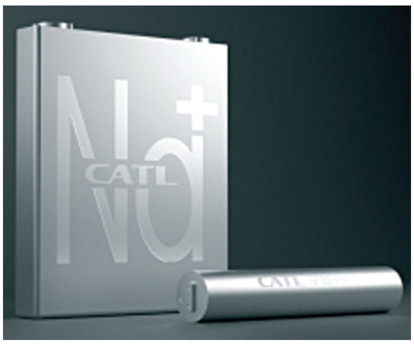
of sodium, which is 23,600 ppm compared to lithium's 20 ppm in the Earth's crust, and the lower cost of extraction and purification. Unlike lithium batteries, sodium batteries retain almost all the charge when temperatures fall below freezing point, resulting in a more sustainable and environmentally friendly performance. However, due to their relatively lower energy density by both weight and volume, SIBs are not suitable for use in electric cars where size and weight are critical parameters. The figure on previous page summarises the key differences between SIBs and LIBs.

The key SIB players in the world

Most major companies manufacturing LIBs are showing an interest in manufacturing SIBs since their existing infrastructure can be adapted with a few alterations. The market size is expected to reach around US$3.2 billion by 2027 based on the growing demand for laptops, mobiles, remote access devices, and other electronic products. Several players, including Contemporary Amperex Technology (CATL), Faradion, HiNa Battery Technology, TIAMAT, Natron Energy, and Altris AB, are active in the field.



SIBs made by CATL (Credit: www.enfsolar.com)

CATL. CATL, the battery manufacturing giant of China and the world's largest manufacturer of LIBs, is going to launch large-scale production of SIBs during 2023 using Prussian blue analogue for the positive electrode and porous carbon for the negative electrode. The batteries will provide a specific energy density of 160Wh/kg and cost less than LIBs. CATL is also developing a compact battery pack that combines sodium cells and lithium cells, leveraging the weather resistance properties of sodium cells with the extended range of lithium cells.

Faradion. Faradion, a UK-based startup working on SIB technology for several years, has been acquired by India's Reliance Industries (RIL). Faradion has developed pouch cells using oxide cathodes, hard carbon anodes, and a liquid electrolyte. The batteries provide an energy density of 160Wh/kg, a cycle life of 300 (100% depth of discharge) to over 1,000 cycles (80% depth of discharge), and can be transported in the shorted state (0V), reducing the risk during movement.

Cylindrical and square SIB cells being produced by HiNa Battery Technology (Credit: https://www.equipment-news.com)

HiNa Battery Technology. HiNa Battery Technology, a China-based company carved out from the Chinese Academy of Sciences (CAS), focuses on the development and production of next-generation SIBs that are longer-lasting, low-cost, and provide better energy density. HiNa has developed three types of SIBs: one with cylindrical cells and two with square cells, with gravimetric energy densities of 140Wh/kg, 145Wh/kg, and 155Wh/kg, respectively. The batteries are based on Na-Fe-Mn-Cu oxide cathodes and anthracite-based carbon anodes. In 2019, HiNa became the first company to install a 100kWh SIB in an electric test car, Sehol E10X, in partnership with Jianghuai Automobile Co (JAC).

Tiamat. Tiamat, a Swedish company, has developed 18650-format cylindrical SIB cells based on polyanionic materials, which are suitable for fast charge and discharge applications such as mobility and stationary energy storage. The batteries provide an energy den-

Different types of SIB pouch cells developed by Faradion (Credit: https://www.bestmag.co.uk)